Howard Marks: AI, Debt vs Equity & The Next 40 Years Of Investing | Nikhil Kamath | People by WTF

Could you also say that indexation and ETFs have taken over because the markets have been in an uptrend? My answer is that indexation has taken over as it has not because it's so good but because active management was so bad. Because a lot of the distressed bonds that you are in is the chance of a downside zero. So I'll give you a little bit of a background.

So most of these are like conversations. Most of the interviews that we do are conversations, they're fairly free flowing and the audience we cater to are entrepreneurs of Indian origin who are around the age of 25 to 30. Everyone's looking to invest in the market, trade the market, start a business, building something from the bottom up.

So we are speaking to them. So most of today is around that. Can we start with a little bit about you and how your own journey began from Queens to Wall Street, maybe like a, how do you describe your story in your words, your story, your origin story? Well, you know, I was born in Queens, New York, as you know, one of the, one of what are called the outer boroughs.

Middle class upbringing. Neither parent went to college. But my father was a very intelligent man and an accountant.

And I think he did his job well. And so we lived a comfortable middle class existence. I went to the public schools of New York at a time when you could get a good education in the public schools of New York.

And I think I got one. And oddly enough, ended up taking courses in business law and then accounting in high school and really connected with accounting, with the orderliness and the symmetry. It just clicked for me.

So I decided to go to business school. I applied to Wharton as the best business school for undergraduates in America. I was told I wouldn't get in, but I did.

Went to Wharton as an accounting student, switched my major to finance, then went on to get an MBA in accounting from the University of Chicago Business School. And between years of business school, I had a job in the investment research department of Citibank, liked it, went back. That's my background.

Is Wharton, would you say, still one of the better schools? No, I think I have no reason to believe it's not still the best undergraduate business school. There are other good graduate schools that are very competitive with Wharton. I'm no expert.

I'm going to say that a hundred times on this tape. But I believe that Wharton is still the best undergraduate business school. And you studied Japanese literature at Wharton? Yes.

When I went to Wharton, there were two very enlightened requirements. You had to have a semester of the literature of a foreign country and you had to have a non-business concentration, what we call a minor. Right.

And, you know, the wonks would take stat or economics or policy. What are the wonks? Wonks? Yeah. Oh, you know, people who just think about making money.

Okay. Or just think about numbers and business. But for my literature requirement, for some reason, I don't remember, I took Japanese studies, Japanese literature.

I think perhaps it was the exoticism. And I just fell in love with it. And so I ended up taking a second semester of Japanese literature and then two semesters of Japanese civilization, then one semester of Japanese art.

And what did that teach you? Well, it taught me a lot. But the main thing I've carried away have been some aspects of the Japanese philosophy. The most important one, the most relevant one is something called Mujo, which literally means the turning of the wheel of the law.

And what it means in everyday life is it means the inevitability of change. That change is inevitable, unpredictable, and uncontrollable. And that we have to accommodate to it and make the most of it rather than think we can control it.

Should one expect it? One should. It's inevitable. Yes.

Of course, one should expect change. One should not expect a specific change because it's unpredictable. I once read somewhere that a Western man, if it rains for five days, will expect it to rain on the sixth.

Whereas an East Asian person, say Japanese, for example, if it rains for five days, will expect no rain on the sixth because the pendulum, like you say, swings back to the mean. Is that culturally true? I have no reason to know. I don't know.

But what I would say is if the probability of rain is, let's say, 50-50, if some people think five days of rain means the sixth will rain because of continuation of trend, and some people think five days of rain means that it won't rain the next day because the rain has all been used up. But I think on the sixth day, it's 50-50. The previous days do not matter for the sixth day.

Well, I mean, and whether that may not be precisely true, but I'm a big believer in the independence of events. And I think that that's what Mudro is about, independence of events. So, you know, if you flip a coin 10 times, better example, coin than rain, because rain, there are some physical properties at work.

But if you flip a coin 10 times and you get 10 heads, the probability of heads on the 11th flip is still 50-50 because the trials are independent. Do you believe in game theory? Well, I believe that you

can strategize and you can figure out which action has the highest expected payoff or the highest maximum potential payoff or the least probability of a really bad payoff. And you can perhaps even strategize about what your opponent is going to do.

I believe in strategy. I don't know if that's exactly the same as game theory, but I think that, you know, as I've suggested in my comments so far, what I really don't believe in is predictions. And I believe that given human nature and the dislike for ambiguity, people pursue predictions so that they can understand what world they're going to be operating in and deal with it accordingly.

And if they do this to excess, then they theorize about a world they'll be operating in which is not as likely to obtain as they think, which means that their predictions can be misleading. And about 20-odd years ago, I wrote a memo called, You Can't Predict, You Can Prepare. And that's really the essence.

And that, you can tell that that comes directly from the understanding of Mojo. We can't predict the future. We can't control the future.

But we can prepare for the future. Now, that raises an interesting contradiction in terms. If you can't predict what the future will be, how can you prepare for it? And the answer is you can prepare for a future which is unpredictable.

So you can, in investing, you can put together a portfolio rather than one that will produce maximum success if a certain future unfolds. You can put together a portfolio which will do fine if several, any of several futures unfold and not too terribly if a bunch of other futures unfold. Now, you can't prepare simultaneously for the positive tail and the negative tail.

If you, if you prepare for a total disaster, then you're unprepared for a great future. If you prepare for enormous success and you're going to maximize your outcome, should this success unfold, then you're probably totally unprepared for a very negative. But you can prepare if you, if you suboptimize for a number of possible outcomes in the middle of the probability distribution.

Isn't that a bit contrarian to the coin example? You said if you toss a coin and it comes heads 10 times, and I toss it the 11th time, you still expect the odds to be 50-50. So I can't look at past data to predict the future or prepare for the future. Then how do I prepare? I think that the, I think that the, the difference is that the coin tosses are independent.

The result of the preceding coin tosses has no impact on the next one. That's not true in the areas I work in. Most importantly, history, the things that have happened so far, and people's reaction to history have semi-predictable implications for the future.

Not history, but people's reactions of it. Well, not history itself as much. I mean, if you, if you knew history, but, but, but, but, I mean, history is the history of what people did and what people did in the past by itself has implications for the things that will unfold tomorrow and next month and next

year.

So in a way, are you saying the psychology of people, mass mentality in a manner remains the same and that can be modeled for more so than events can be? Well, of course, events are largely the things that people do, which is largely the result of their psychology. But I do think that, that events in the future are somewhat predictable because, because the implications of past behavior are somewhat predictable and future behavior is somewhat predictable. The things that have happened, for example, in the economy and the markets are likely to induce certain behavior in the future.

And you can know a little bit about that. And, but when, when the history and the past behavior has been extreme, the ability to infer from that rises. In other words, I wrote a book about cycles and that's the one where I refer to my experience traveling in India and cycles are induced primarily by behavior.

By, you know, the upcycle is exaggerating, overgeneralizing. The upcycle, there's a trend line. Let's say it's GDP growth.

There's a trend line, but sometimes the economy grows more than that. Why? Largely, I think, because of optimism. Because people are optimistic, so they spend a lot of money.

Producers are optimistic about demand for their products, so they build new factories and hire labor and buy machinery. And the sum of these things produces a period of above-trend growth. But then, when producers have more than prepared for the coming growth and consumers have been sated by consumption, then things will ease off and maybe it results in a period of below-average growth.

So, you know, I wrote the book. I was about two-thirds through writing the book and I've been studying cycles for almost 60 years now, living through cycles. And I said to myself, why do we have cycles? Why doesn't GDP, if the average growth in GDP is two, why isn't it always two? Why is it three or four or one or zero or minus one? And the answer is excesses and corrections.

And then you go to a negative, which is an excess, and then it corrects back to the trend line. So rather than thinking of cycles as ups and downs, which I think most people do, think of them as excesses and corrections, excesses and corrections, fluctuations around the trend line. I mean, this is most easily seen in the stock market, for example.

You look at the S&P 500. The S&P 500 has returned about 10% a year on average for the last 100 years. But the constituents of the S&P have changed many times in that period.

But the average rate of return doesn't change. Average rate of return. I don't know about the S&P.

I know in India, the indices, I'm a stock investor. That's my full time job. We have returned about

11% over the last couple of decades.

But we have depreciation of currency, which is about 4-5% a year. So I'm assuming the S&P would have returned 6% if you were to ask me to wager a guess. But I don't think it's a fair benchmark because the companies in the S&P, like the indexes back home, change by virtue of market cap.

They do. But my point is that the S&P, however constituted, has returned an average of 10% a year for the last 100 years. But not steadily.

In the 90s, it returned 20% a year. And in the aughts, it returned zero per year. So certainly not steady.

And not only is that an interesting phenomenon to think about, but even more so, the fact that the return on the S&P, which averages 10, is almost never between 8 and 12. This is a very interesting datum. Why is it that the norm is not the average? Greed and fear.

Excess and correction. Excess of optimism, correction. Excess of pessimism.

Can I ask you a small digression in a question? If humans were no longer in the picture, and AI models are trading in the market, would 10% be 10% every year because there would be no optimism and pessimism? This is another of those times when I'm going to say I'm not smart enough to know. It stands to reason that it would. If we programmed AI to find the elements and expect the patterns that produced success in the past and left it alone, and since we're told AI does not have greed or fear or optimism or pessimism, you would think that the returns would be steadier.

I think it makes sense to think that. I heard your interview a year ago with Nikolai from the Norwegian Sovereign Fund. He's a friend of mine.

And when he asked you, where are we in the cycle, you said somewhere near the middle. Well, I think that was two years ago. Where are we now? Well, guess what? It's two years later.

That's the only thing I know for sure. So I believe that interview was April of 2004. And the market continued upward from April 24 through, I think it's fair to say, January 26.

So another 21 months took it considerably higher. And of course, the economy, when you look at the economy, people think of the age of the recovery. So the economic recovery is two years old.

The bull market in stocks went on another almost two years. So I would say that the, you see, first of all, as the economy, it's very, very hard to date or talk about the age of this cycle because we had the pandemic where we did something unique. We closed the world economy to reduce the spread of the disease.

And so that interfered with the normal workings. And according to the people who make these decisions, it produced a recession. It was a very unusual recession because it was only one quarter.

I always thought a recession was two quarters of negative GDP growth. And of course, it didn't occur for the normal reason of excess optimism leading to a correction. It was manmade and was readily corrected.

The second quarter of 2020 was the worst quarter in the history of the world for GDP. And then the third quarter was the best. So was it really a recession in the functional terms of determining cyclicality or or just some noise? And can I ask another question here? I would say GDP, if I were to consider it the total produce going into the economy and consumers consume as much as we can.

I could argue that a recession is truly when there is more production in an economy and the consumption is organically lesser. So do you think GDP is a good benchmark to figure out when there is a recession or not? Because the consumers went away then artificially. Well, I think it's I think I think your definition of recession isn't isn't right because what you said was there's more production, but that's not less consumption, less consumption.

That is that's the main element of of of a recession, in my opinion. Manufacturers are always happy to produce goods to meet demand. The recession, it means primarily a reduction of demand so that which which discourages manufacturers from producing and the reduced production is really what causes the recession.

Fewer people are at work, etc. Fewer goods are produced and consumed. So in the pandemic, production went down, but the demand for consumption did not actually go down, but just that the markets were shut for a period of time because when they opened, people really consumed.

Well, I mean, yes, the the underlying demand didn't go down, but people I mean. In the sense that people would have liked to have parties, they just weren't allowed to have parties. So the demand, the psychological demand for a party was there, but the economic demand for rental halls and bands and food and party dresses and things like that didn't exist.

So that's that. So but this is why I say that the GDP was kind of artificially depressed by a decision of government, really. So so when we say that how old is this recovery? Was that a real recession? If that was a recession, then this recovery is now six years old.

If that wasn't a recession, then this recovery is 17 years old, which is the longest in history is 10 years. Why 70? Because the last recession was in 09. So it's 17 years since then.

But can you just tie recessions and cycles down to tenure time? You can't. And that's one of the reasons why I don't believe in predictions. But there seems to be a norm.

We always said that the average was eight years for for a recovery. And in the 2010s, it went on for 10 plus years. So we called that the longest in history.

So so and and normally, if we have a recession in year X, we don't expect another recession two years later because recovery goes on for some period of years, which seems to average eight. I can't explain why. I don't think anybody can tell you why.

Maybe maybe the answer is on average, eight years is long enough from the last one to prevent to permit the next one. But so I don't believe you can. You know, somebody I wrote the book about cycles and somebody in England said, oh, no, these aren't cycles because they're not predictable.

You know, cycles like radio waves or sine waves or something like that. These are predictable. And and I think that it patterns in the in the in the real world, the non non scientific, non mechanical world certainly aren't regular, but they do occur ups and downs.

They do it predictably, but not necessarily predictable in time or predictable in amplitude. But they certainly occur predictably. So if I can bring you back, Howard, to the question.

What question is so long ago? Where are we in the cycle now? Well, I think I think that the the recovery is. I think it's it's gone on a good while and it's it's not nascent. It's not adolescent.

It's but it's also not geriatric. I think it's in middle age. And and I think I think the U.S. economy, if that's what you're talking about, is healthy and and doing quite well.

If if the thing that presages a downturn is excesses to the upside, you don't find too many of those. You know, when I go to a city, I look for construction cranes in the sky to tell me if there's a building boom. Most booms are followed at some point by busts.

I don't see it in most of our cities. I don't see inventories growing. Inventory growth is.

Inventories grow for two reasons. In times of weak consumption because people don't buy the things that manufacturers produce. In times of strong consumption, inventories grow because manufacturers amp up production to get ahead of some anticipated surge in demand.

And those often presage downturns when they turn out not to be warranted. I don't see that. In some sectors, do you? Well, I mean, maybe I don't I don't look at the data finely enough to tell you that.

In AI, maybe. Pardon me? In AI, maybe. Well, of course, we don't talk about inventory in AI, but there's certainly a boom in capital investment for AI.

One could say hardware, energy, computers, inventory. Yeah. Well, I mean, look, there's massive construction of infrastructure for AI, much like buildings.

Yes. Yes. Yes.

That's right. Well, that is that is the area that's booming in our economy. And I need people who know more about AI than me to tell me whether it's going to bust.

You know, the question is, is all of this or is some of this construction unwarranted? I don't think anybody can tell us that. We do know. We certainly know that it is an area of a boom.

How do you stay so sharp, Howard, at your age? Like, I have watched a whole bunch of your interviews and read a lot of what you've written over the years. As a stock market investor all my life, I think there's so much I can learn from you. But how do you stay this sharp today? Well, I don't think I don't.

I mean, I'm lucky. I have. Genetically? Well, I think I'm I think I'm very lucky genetically.

OK. Yes. Because my dad lived to 101 and he was quite sharp at the end.

And, you know, we all face a decline at some point in time. And the only question is, when does it start and at what rate does it progress? And, you know, look at Warren Buffett, Charlie Munger, some of my role models and friends. And, you know, they were or are sharp in their mid 90s and late 90s.

So it's just I mean, I mean, when you say, how do you stay sharp? I don't go to the gym and work. Well, I do do a lot of puzzles and I read and I read in fields that are not my fields. I you know, I'm still trying to gain knowledge and broaden myself.

And I guess I make some effort to to plow new ground. And the question is, when do you stop doing that? At what age? But, you know, my last two memos. I hope your audience knows about the memos.

Yeah. But I wrote one December 9th and one, I think, February 25th. About AI.

And I had to do a lot of learning. You changed your mind on AI. I changed my mind.

But first I first I challenged my mind. And, you know, I had for the second one in particular, I had an extensive conversation with Claude and and learned a great deal. And, you know, it was very exciting to be plowing new ground at almost 80.

And do you think it could be ambition and hunger? Well, it's it's it is definitely a hunger. It's a hunger to learn and to stay relevant and and to stay stimulated. When you say ambition, that that usually starts with dollar signs.

Not necessarily. Yes. Well, it's it's an ambition to stay relevant and and and to to use your terminology to stay sharp rather than start to atrophy.

I think staying relevant is the core ambition of everyone. The dollar is a means that people pick in the beginning, not at the end, maybe. Well, I think I think that most ambition starts with the dollar sign.

In enlightened cases, it transitions to non-dollar. Not every case. You know, there are people are different.

You can't. Mark Twain, I think it was, said all generalizations are flawed, including including this one. But you can't generalize.

Like you say very well, I heard you speak about the role luck plays in life. Would you say enlightened or lucky? The ones who transition. Well, now now you're really getting deep because now you're talking about determinism versus intention.

And I think there are some of both. I mean, some people kind of I mean, these things are hard to parse, Nikhil. And but some people, I think, transition naturally.

And as they grow, maybe they become wiser and they start to understand that there are things that are more important than money or, let's say, things that are important in addition to money. And so they do it. Let's say that naturally, not intentionally.

I imagine there are people who do it intentionally. People who go to the mountain and meditate and think about it and conclude, you know, I really should change my focus at this point in my life. Have you been to have you been able to ever pull that off? Like I've been fascinated with the idea of going to a mountain and arriving at some kind of evolutionary thought in my mind, which makes me actively transition into something.

Have you been able to touch that? Well, I haven't physically gone to the but I think the key, I've written the first pages of a book. I think the key to all of this, to all of life, is to behave thoughtfully. With your mind engaged, not just let the river take you, but give thought to what's going on.

Why is it going on? What does it mean? Why did it happen? What does it imply for me? What should I do about it? And that is, I think, going through life with your eyes open. And that's how you can intentionally get to a higher stage in life by doing that. And the image that comes back to me is the image of whether you let the river take you.

Or whether you try to figure out a better place to get to and then try to get there. Maybe once or twice in my life, I have felt like I could see the river. See the river, yes.

Can you actively? I think at this stage I am able to think about this progression constructively. When I think about my early decades, not years, decades, and I think about, you know, what I did from, let's say, I would say from the beginning of, let's say, high school, 1960, until starting Oaktree in 95. So that 35 year period, which took me up to age, let's say 49.

I was adrift. I did not make proactive decisions. I let the river take me.

I'm very lucky that it took me to some good places in several different aspects of life. But I don't consider, when I look at my behavior, I don't consider it thoughtful or intentional. And so I'm just lucky that I got to a good place despite my own inattention.

I think that starting Oaktree was the biggest and maybe first thing I really did with intention. And I've tried to be more intentional since then. But when I look back and I describe how I selected my career, how I selected my first job, how I decided to go to graduate school, and how I transitioned from the equities department to the bond world in 1978, in time to in time to have the benefit of the birth of high-yield bonds.

It was all happenstance, serendipity, coincidence, passivity. You know, I was not making intentional decisions in that period. What changed? How did you become intentional from being unintentional when you began Oaktree? Because other people can maybe learn from it.

Maybe I can learn how to catch it. Well, the easy answer is that my wife pushed me to think about it more and to independently start Oaktree. My partner Bruce Karsh and I did it together.

So his participation encouraged mine. But that required a proactive decision. You know, the biggest change I made before that professionally was moving from Citibank to TCW in 85.

They approached me. That was not proactive on my part. Not the act of starting Oaktree, but from being adrift the river to seeing the river.

Well, I think once I made the decision to participate in the starting of Oaktree, I couldn't drift anymore because now I was the person leading Oaktree. I had to make proactive decisions. I mean, it became my job.

And maybe I warmed to it. Maybe having made the proactive decision to start Oaktree and having done it, maybe I said, hey, this is good. I think that maybe I like leadership.

And leadership is by definition proactive. It seems like an oxymoron to say he led passively. Those two words are in opposition.

So I think that, thank God I rose to the occasion and left TCW with Bruce and my other partners to start Oaktree. And then I think that it was, I don't think I ever said, well, now I'm going to start taking control of my life. Now I'm going to start acting intentionally.

Now I'm going to become a leader. I just think it was inescapable. That's very interesting because it sounds like you're saying entrepreneurship and taking risk can trigger the riding the river to seeing the river.

Some people are driven entrepreneurially. I think that entrepreneurship is the epitome of

intentionality. It's taking the bull by the horns, as we say, proactively.

So I think the entrepreneur naturally, innately does the opposite of drifting down the river. And my inclination was not entrepreneurial. I liked most of what I was doing at TCW, but with encouragement from Bruce Karsh and my wife, and with a little encouragement from TCW to leave in terms of how I thought I was treated, my inertia was overcome.

That's what I would describe. It wasn't a proactive decision. Hey, let's get going.

It was the unlikely overcoming of my innate inertia. It's a very different story. It could be like a chicken and egg thing.

Oh, it's very chicken and egg. Yes. But I think the important thing is, for the purposes of your audience, that my departure from TCW did not start with an entrepreneurial spirit.

And some people, I know people who always dreamed of making a lot of money, always dreamed of running their own business, chafed at being an employee, had to get out and start their own. That was never descriptive of me. So I kind of did something that you might call entrepreneurial, despite myself, is the way I would describe it, rather than because of my innate drive.

So Howard, my job for 20 years has been that of a stock investor. You've been a debt person almost all your life. Why would you be in an asset class which has finite upside? And the downside, I guess, is finite, because in most cases, it's zero versus equity, where you could have exponential upside.

Well, that's a great question. And I haven't really been asked that question much in the past. My answer may take a while.

Settle in. The question, as you pose it, is an interesting question if you assume rationality and objectivity. But that ignores personal, let's say, kinks.

So I'll return to that. But first of all, my decision to move from, I spent nine years in the equity department, so it wasn't really at the very beginning. It felt like a long time.

But my decision to move from the equity department in 69 to the bond department in 78 was not my decision. And it wasn't because I figured out it would be better for me. The city bank I joined in September of 69 was an investor in what was called the Nifty Fifty, the 50 supposedly best and fastest growing companies in America, companies where nothing bad could ever happen.

There was no price too high for the stock. And if you bought those stocks the day I got to work in September of 69, if you held them tenaciously for five years, you lost about 95 percent of your money, 95 percent. Because for many of them, something did go wrong.

And for all of them, the price was too high. The P.E. ratios, as I recall, were mostly between 60 and 90. So that was a real disaster for the people who invested in the Nifty Fifty, which was most of the money center banks.

And I was director of research by the mid-seventies, so I was part of the process. And we hired a new CIO who wanted to have a new head of research. And I helped him hire my successor.

And then he said, what do you want to do next? I said, I don't know. I could do this. I could do that.

Typically, I didn't think it over very much or assess my strengths and weaknesses very well. And he said, I'd like you to move to the bond department and start a convertible bond fund because he had come from a place that had had one and it was very successful and we didn't have one. And I said, yes.

So that was the extent of my intentionality, which is almost non-existent. It was, you know, and I say now and it may be an exaggeration, but I was lucky I didn't get fired. But American companies gave pretty much lifetime employment at that time.

Anyway, so number one, my move to the bond department was not voluntary, other than I could have said no. But number two, you talk about the fact that bonds have capped upside and some downside from default. Certainly true.

But it fit me because I was brought up to be very conservative. My parents were adults during the depression. Now, your parents probably weren't alive during the depression.

And most people I know who are old, their parents were children during the depression. My parents were adults during the depression, which meant when I grew up, what did I hear? Don't put all your eggs in one basket, save for a rainy day, avoid risk, et cetera. So I think Indians, Indian parents live through pseudo-socialism in the country.

Mm hmm. Yeah. Not as bad.

Yeah. Yeah. But, but, see, you as a stock guy, look at bonds and say no upside, some downside.

I say predictable outcome, which is achieved almost every time. Is that true, though? Because a lot of the distressed bonds that you are in, which have fairly high coupons, is the chance of a downside zero? I said almost every time. And, you know, first of all, so as I said, I moved to the bond department in 78, May of 78.

In August of 78, I got the phone call that changed my life from the head of the bond department who said there's some guy in California named Milken or something who deals in something called high yield bonds. Do you think you can figure out what that is? Because the client had asked for a portfolio. Junk bonds.

Pardon me? Junk bonds. Milken, Michael Milken junk bonds? Yeah, junk bonds. So, so I said, yes, I can do it.

And that put me here today, 48 years later. But, but, so I, if you've read Malcolm Gladwell, Outliers, you know the importance of right time, right place. I was in the right place at the right time.

I was in, that put me in high yield bonds at the beginning of the high yield bond movement. I've met Michael Milken and because he got unlucky later in life, it couldn't have just been right place, right time because it didn't go that way for him for a period of time. Well, that's right.

But things in his existence conspired to produce a bad result for a while. But not for me. I got there at the beginning.

What Malcolm Gladwell's book is really about, Mikheil, is about it's great to be at the front of the line. That's really what it's about. He talks about demographic luck, right? Right time, right place.

I think the question is, how do you figure out where is the front of the line? Well, I didn't. Somebody said, would you please stand on that line? And I looked around and I was the first person in the line. You see? So that's luck.

Figuring out is much harder. And by the way, not all the people who were first in the line were there because they figured out that that was the line to be on. A lot of it is serendipity, you know? And not all of them did well either.

That's right. That's right. Not all of them did well.

But anyway, the point is, I've been in high-yield bonds for 48 years. And in our experience, 99% of the bonds have paid interest in principle as promised. So I think I can say almost every time.

And if you earn on average 10%, you only needed to be right 90% of the time. No, that's not right. Because if 10% of your bonds defaulted and you lost 10% of principle and you made 10% of interest on the ones that paid, you had a zero return.

You had to reconstrain your losses to a much lower figure than 10%. You had to be right 97, 8, 9% of the time. And we were right 99% of the time because I think we did it in an above average way.

But the point is, that reliability appealed to me. And I'm not a futurist or an optimist. So I was well suited to it.

And then, of course, we got into the distressed debt business when Bruce Karsh joined me in 87. We brought out what I believe was the first distressed debt fund from a mainstream institution, or one of them. And he runs those funds.

He's managed about 70-odd billion dollars since 1988 in that field, by far the biggest. And of his total profits and losses, well over 90% are profits, less than 10% are losses. So again, I think you can say most of the time.

And so the regularity, the contractual nature of returns on bonds appealed to me. And so this is why I've been very comfortable in that field. The other thing is, frankly, a lot of your success in investing is determined by other people.

If you go into a field that everybody likes and they like it a lot and have bid up the price, then your returns are not likely to be high. If you go into a field where everybody else has been blind to the merits and says, I wouldn't touch that with a 10-foot pole, and you figure out that it's good and it turns out to be good, that's how you get an exceptional return with low risk. So I think that for the most part, well, I was lucky at, I should say, at the time I made these decisions actively or passively, took these steps, shall we say, I went into fields other people didn't like.

You used the term junk bonds, a derogatory term. Nobody talks about junk stocks, right? But I mean, junk bonds are much more predictable than stocks, and yet they called them junk. We call them distress bonds.

Yes, but that was a bias that made them available to me cheap. And that was a very good thing. And I could see that and take advantage of it.

Can you contextualize distress bonds? Let's say the US Treasury, US government borrows at 4%. Let's say I read somewhere that Google is borrowing 100-year paper at 6%, which sounds crazy, and we'll get to that. What kind of companies are in this distressed bond ecosystem and what rate of return are they paying? Rate of interest? Well, let's, we have to clarify.

Oaktree and I have two businesses. One we perform, we call performing credit, and one we call opportunistic credit. That's the new euphemism for distressed.

So wouldn't you benefit from calling it junk so you have a higher margin because more people would stay out of it? Yes, if they listen to me, if I was that influential. And that's the right kind of counter-thinking. But on the other hand, a big part of our business now is making loans to companies that need what we call rescue loans.

They don't like to be associated with the word distress. So we're more likely to get their business if we call it opportunistic. So, you know, in the performing credit area, we lend money to companies that the world thinks has a, let's say, 3, 4, 5, 6% of not paying us.

3, 4, 5, 6% probability of not paying us. We lend them money if we think it's one or two percent. So if the world thinks they're 5% likely to not pay us, and we conclude that they're only 2% likely to not pay us, then the world requires them to pay a rate of interest which is commensurate with their 5% probability of non-payment, which means if we're right and it's only two, we're getting

something for nothing.

And that thing is called excess return. So that's what we do. That's the essence of lending money.

I mean, why would you lend money to a company that has a non-zero probability of paying you back? And the answer, Mike Milken's answer, was you can demand a rate of interest which is compensatory or more than compensatory. And when you get, when you get payment in life for doing something which is more than compensatory, that's called excess return. So is the skill set at Oaktree calculating that spread? And how do you do that? Well, when you say the spread, the skill set at Oaktree, the base, the most important fundamental skill set is predicting the probability of default better than other people.

That's where, if we don't have that, that's the necessary condition for superior performance. And is this skill set company specific or are you able to call the larger cycle in the market and hence you do it better? No, that skill set is separate from the latter thing you talked about is what we call macro. That skill set is not, is micro.

We have great analysts who look at companies. We have a framework for analysis. The framework is essential, necessary, but not sufficient.

What makes it sufficient is having brilliant people to operate the procedure. The procedure itself is nothing without superior implementation. But we have both.

And so over the last 40 years, on average, something like 3.6 or 3.7 percent of all high yield bonds have gone into default every year. And our default rate has been roughly a third. That's our superiority.

And that is a case by case bottom up superior implementation. And when you say superiority, not the process of the people running the process, do you mean attention to detail? Do you mean thinking out of the box? No. Why do we have a lower default rate? Well, I think we have institutionally more experience than anybody else.

We've been doing it longer. We have a a a work environment that allows people to stay in that job for their lifetime. You know, when I started at Citibank in 69, every analyst goal was to become a portfolio manager because that's where the that's where the luster was.

So so and as long as that's the case, then on the analytical side, you don't build up institutional excellence because it's people are always trying to get out of that job. You want people to build up excellence and stay in the job so that you can get the benefit of their excellence. And and we set up that system.

So, you know, you can be a lifetime analyst and be very successful if you're great at it. So that's one important thing. So we have people who've been doing it for years and years, decades, and we have the benefit of their expertise.

We have the institutional experience, which gives us stability in times of rocky psychology. When people get too excited or too depressed, we tend not to. That's a great help.

We have a a process which enumerates a eight areas of investigation and many subsidiary questions within that. And everybody has to follow it. Now, people might say when I when I started doing something like this at Citibank as director of research, people said, oh, you're taking away our creativity.

But we we we told them that a process they had to follow, but they could follow it creatively. But the process consisted of covering the basis. You got to cover the basis consistently.

You got to ask the same questions of every company every time. Rather than, you know, in in the equity world where you live, you'll get a research report. This is by this company.

Great management doesn't talk about the product by this company. Great product doesn't talk about the management by this company. Great tax shelter doesn't talk about the product or the management.

I felt that it was important to cover every base every time. And when when you when you look at a at a portfolio and you see a stock that's down 80 percent, it's usually because something wasn't covered. And because the the the analysis was not disciplined enough to touch all the bases.

And again, you know, you're talking about the difference between stocks and bonds. That's one of the differences, right? We're we're we're bean counters. We're green eye shade types.

But at least we should do it well and consistently. The equity investor has imagination and foresight and entrepreneurial spirit and but doesn't do it as consistently as we do our analysis. If you had to begin a new career today for the next 40 years.

Yes. Would you pick debt or equity again for me or for or for the average watcher? Let's say for the average watcher. Well, I think that I think that we were very fortunate in getting into some debt markets before they were discovered and before people understood them.

And we benefited from people's antagonism toward those markets. Those opportunities are in the past. So where is the front of the line today? Well, I mean, who is the person in the investment business today who will find the most success in the next 10 years? The answer is, in my opinion, the person who best understands A.I. and its capabilities and implications.

So, but. Or is a contrarian and doesn't believe in A.I. Pardon me? Or he could be a contrarian and bet against A.I. Well, that's what I said. I said best understands.

I didn't say is most in favor of. So if if A.I. is going to disappoint, the expectations are very high. And when you as you know, from being a stock picker, when you go into something where the

expectations are very high, it's very easy to lose money if those expectations aren't rewarded.

So, yes, maybe maybe if you're the person who best understands A.I. today, maybe you'll make a lot of money by understanding that it's overdone and betting against it. The current expectation with A.I. is almost dystopia. Or utopia.

Yes. Which is crazy, I think. Well.

I don't know if you read my memos, especially the last one, you know that I was shocked by my experience with A.I. and Claude and what it. But you also said you haven't fired anyone, nor do you intend to because of A.I. I believe. I don't know enough to be confident in this, but.

I believe. That A.I.'s excellence is in. Discovering past patterns, extrapolating them and applying them with discipline and with.

Let's say calculations or logic, which is almost always correct and not subject to psychological ups and downs. But I think there's more to investment excellence than that. It's the innovation of new patterns.

It's seeing the potential of things that have never, you know, how if what A.I. does is pattern matching, shall we say? Pattern completion from. Yes. Then how will it deal with something where there's no pattern? Something that's brand new.

And, you know, I wrote a memo. Is there such a thing? Well, look, I wrote a memo. Nine or ten years ago called investment without people investing without people.

It was pretty early days, so it had three levels. Indexation and passive investing. Algorithmic or systematic investing.

A.I. and machine learning. And my revisionist memory. Or I should say my my memory, which could be revisionist, tells me that in that memo I said I posed some questions.

I may not have posed them, but I wish I had if I didn't. Can A.I. sit down with five business plans and figure out which one is Amazon? Mm hmm. Can A.I. sit down with five CEOs and figure out which one is Steve Jobs? That goes beyond pattern recognition.

I would bet not. Well, I would bet not too. So if that's true, if we're right, that means there's still a role for people.

And but before you go on to the next question, I have to interject something. Neither can most people. Mm hmm.

So so what that says is there is a role for exceptional people. Even in an A.I. world. And that's my conclusion.

I hope I'm right. I hope to be one of them. I hope my firm is is is still one of them.

But if that's true, if there are things that A.I. can't do that some people can, then that's what I want to keep doing. Now, you look, you're a stock picker. When I went to graduate school at University of Chicago in, let's say, 66, no, no, 68, the professor said most mutual funds underperform the S&P.

So and they charge high fees. So why don't they just buy one share of every stock in the S&P? There were no index funds or concept of indexation that came along in 74 primarily with Jack Bogle. And now.

Most mutual fund equity capital is run by passive or index. Why? Low fees. Is that the only reason? No.

It also the active management didn't work. They were unsuccessful and charged high fees for it. But even if they had low fees.

Even if they charge the same fee as the index fund, if the passive decisions are inferior, indexation is still better than active. So it's not the low fees that was that exact that exacerbated the problem. It was really.

So my answer is that indexation has taken over as it has not because it's so good. But because active management was so bad and but I believe that people can do some things that I can't. Could you also say that indexation and ETFs have taken over because the markets have been in an uptrend largely in a more volatile time? Active might make sense.

In a depression, active might make sense. You know what? The bad times. Create an opening for active management, but it's still hard and not many people can do it well.

Why do bad times create an opening? Because panic drives down let's say stock prices. But the same panic causes most people to panic, which means that most people can't stop up and buy in the panic. And there's a there was a guy named Wally Deamer who was an old time trader who had some great quotes and he turned them into a little book.

But his greatest quote of all is when the time comes to buy, you won't want to. So this this idea that upheaval creates opportunity. Is logically correct, but not realistic because most people's psychological impediments prevent them from taking advantage of the upheaval.

You see, maybe you mix active plus AI together to try and take emotion out of it. Well, that could work. That could work.

If Claude could call you or send you a message which says this is one of those opportunities we talked about. Don't be stupid. Get your ass in gear and buy something.

Don't hide in your hole. That could that could be a good thing. Yes.

But but I still think that I still think, Nikhil, that when you get into areas which are not connected, what I would call mechanically, where physical laws are not everything, that means that human psychology plays a role in developments. And that tells me that people who are superior at dealing with human psychology can get an advantage. And I think that includes investing.

But not everybody. It always comes back to not everybody. And, you know, there's a there's a phrase in in our world.

A friend of mine in England used it as the title for a book. Simple, but not easy. Our job is simple.

We've got to find the best managers, the best companies, the best ideas, the best products. My son is a venture capitalist. He does a great job.

He finds the best founders. The task is easy. I just said it to you in 10 seconds.

It's just not easy because what makes you superior at finding founders? What makes the better founders available to you rather than somebody else? And Charlie Munger once said to me, putting together simple, but not easy. Charlie, when I finished my first book, the most important thing, I had lunch with Charlie. And as I got up to leave, he said, just remember, none of this is easy.

Anybody who thinks it's easy is stupid. That was one of the great things anybody ever said to me, because Charlie said it in his typically brusque way. But it can't be easy to be smarter and more disciplined than all the other smart people who are trying.

That's what the efficient market hypothesis says. I've never seen it written down that way. But all these smart, highly motivated, educated, numerate, computer literate, interwired people are trying to get rich.

So where's the $10 bill nobody's picked up right now? Well, I think it's around AI. But everybody's excited about AI. So if your excitement level, and by that I mean your insight level, is average, you don't have an edge.

And you get involved in AI, your performance will be average. You will go along with the tide if it works. You'll be schmeissed if it doesn't work.

Superior, it all comes down. You boil down our conversation to this point. It all, and the things Charlie said and so forth, it all comes down to superior insight.

And we have an author in America, a guy named Garrison Keillor. And he wrote a book called Lake Wobegon. And Lake Wobegon was a fictional community in, I think, Wisconsin, if I'm not mistaken.

And he said in Lake Wobegon, all the children are above average. Well, we know that there is no place where all the children or all the investors are above average. So I think my own vision of the world, which is not a scientific vision, is that in fields where human nature is involved, and the future is unpredictable, and things don't operate according to mechanical rules, there is still a place for superior insight.

But you will not have success without superior insight. Howard, most of my investing is in India. I'm 39 years old now.

I know you've deployed a certain amount of capital, I think $4 billion to India recently, but you haven't written a memo about it. My assumption is India is growing at say 7% or 6%. It will continue to do so for a period of time.

GDP per capita will go up. So consumer will do well. Energy will do well.

Energy consumption goes up as GDP per capita goes up, maybe above $5,000. Would I be okay in just staying allocated to equity, say 80% for the next 20-30 years? And if so, is there a sector that you like? Well, I've been cautioning you about things I don't know much about. Here I can be more emphatic.

I don't have any idea what the best sector in India is. I will not hold myself out as knowing anything about the Indian stock market. Okay.

I read that during COVID, you lived with your son Andrew in a house. Yes. I'm sure Andrew has learned so much from you.

Is there one thing that you learned from Andrew? Oh, I've learned so much. And this is how we started the conversation. You said to me, how do you stay sharp? And speaking with young people, if I speak to my contemporaries, how am I ever going to learn anything? Speaking with young people is how you learn.

And they know stuff you don't know because they are still learning and they learned more recently. And hopefully you have something you can give them in exchange, which is experience that they don't have because they haven't lived 80 years. But Andrew is essentially your age.

And I wrote a memo after that experience. Three generations living under one roof was of great value. And I wrote a memo called something of value.

And the other reason I chose that title is because we mostly talked about something called value investing. But I get so much from him and he pushes me to see things that I don't wouldn't otherwise see. And he's also intolerant of my not staying sharp and moving ahead.

You know, how do you keep moving ahead at 80? That's really the question. But I don't think it's impossible. But a great example, I mean, the example that immediately comes to mind, and I

think it's a great one, is that he observed and I wrote in the memo, the memo was something of value.

January of 21, as I recall, he pointed out, it was almost revelatory. Readily available quantitative information about the present cannot hold the key to success. Because everybody has it.

Success in investing is doing better than others. Investing is a is a funny field because it's really easy to be average and it's really hard to be above average. But readily available quantitative information about the present is not going to make you above average for the simple reason that everybody has it.

And and I talk in the memo about that. One of the reasons Buffett was able to steal the march on everybody else and become Buffett is because nobody he he he sorted out and he worked it when nobody else did. And, you know, when I was starting off in this business, in the research world, think about your process.

Let's say I said, you know, I went into a store today, I bought a I like the product. I think maybe that company has a future. I'd like to learn about it.

How would you get information? Think about what you would do if you had that experience this morning here in New York City. What did we do then? You know what we did then? We wrote a letter to the company and asked it for a copy of the annual report. It took a week for them to get that letter.

They put it at the bottom of the pile and they took the letter off the top of the pile and they sent that person an annual report. It took a week or two to get down to my letter. And then they took an annual report, put it in an envelope, addressed it to me, put a stamp on, put it in the mailbox, and it took a week for it to get to me.

Now I exaggerate, but the point is it took a month to get an annual report so that I could start studying the company. There was no online. There was no wiki or whatever source you might use.

The only alternative was there were there were books called Moody's manuals and they were, each book was this thick, no this thick, take it off the shelf. It had a paper called onion skin, the thinnest of paper, and it had tiny print, tiny print. You needed glasses and they had the financials, the current financials on every company.

Not much narrative, nothing, no discussion of the future, but just the financials of every company. That's all you could get until your annual report came in the mail. Then maybe you could start figuring out the business and its potential.

If you were Buffett and if you sat in your office in Omaha and read the Moody's manual when nobody else was willing because it was stultifying, you could get an advantage. Is that what he

did? I think, well, that's my illustration of what he did. Maybe he did more.

I don't know. I'm not going to pay you a compliment, Howard, but if I could. Compliments are always a gift.

I think the nimbleness and the willingness you have to change at 80 is by far the most impressive thing about you. When I speak to really successful people, I think this trait over all others kind of makes them who they are. Well, I appreciate that comment.

I hope it's true, but it makes sense, doesn't it? Because the ability to change your mind and change your thinking and learn new things is an example or a component of superior insight, which is what you have to be. This is a competitive game. It's like golf.

You say, today I shot 74. I tell you I shot 74. You have no idea if I won or lost.

The only thing that matters is what did everybody else shoot, right? If I tell you last year I made 13% in my stock portfolio, is that good or bad? Well, it's pretty good. The average return on the S&P is 10, but of course the S&P was up 18 last year, so 13 stank. This is a competitive game.

You don't have to. It's not a matter of being right or wrong. It's a matter of being more right than the other person or less wrong.

Are you able to change because you have been able to consciously remove ego? Well, I think that helps because ego, if you have too much and the wrong ego, would probably, I would imagine, tend to make somebody say, I've always done it this way. It's worked in the past. I'm an enormous success.

I'm worth this much money, so why should I change my approach? But you have to keep changing. Innovate or die. What could trigger it? If we were to advise people watching this, if you were to, how can they trigger the realization that they need to constantly change? Well, it's a funny thing.

This has been a theme for me for the last nine months. I say this every time I get a question like yours, Nikhil. The really hard questions are the ones that start with the word how.

How can I learn the need to evolve and grow? How can I learn to see things better, more clearly than other people? How can I learn to be a credit analyst who better understands the default probability than anybody else? How can I become what I call a second-level thinker, understanding things at a higher level and better than other people? And all the hows, I can't tell you. How can I remain sharp at 80? I can't tell you. I can tell you what you have to do.

You have to do all the things that I mentioned in order to be a superior investor. I just can't tell you how to do it. And in my first book, it's funny, when I had the idea for the first book, which is called The Most Important Thing, it has 21 chapters.

Each chapter says, the most important thing is, and then it's a different thing, because there are so many things that are important. And I told Columbia about the book, and they wanted to publish the book at Columbia University. And so they said, give us a sample chapter.

And I wrote out a chapter, each chapter says, the most important thing is, and this one says, second-level thinking. And I wrote the chapter about second-level thinking. I had never really thought about it before.

It just kind of struck me, and I wrote it down. And what it says is, it's pretty simple. If you think the same as everybody else, you'll act the same as everybody else.

If you act the same as everybody else, your investments will perform the same as everybody else. So to perform better, you have to deviate from the herd. Seems clear.

How do you do it? How do you do it? I can tell you the importance of second-level thinking, but asking me how to become a second-level thinker is like, in basketball we say, you can't coach height. All the coaching in the world will not make your team taller. So I can tell you that you must become a second-level thinker to be successful as an investor.

I just can't tell you how. If I were to draw an extrapolation, that second-level thinking is also contrarian thinking in a way. Do you think AIs which are predictive in nature can be modeled to be contradictory tomorrow? But that's the challenge, you see, because second-level thinking, or contrarian thinking, is not just different from the herd.

That's not enough. It has to be different from the herd and better. So you can say, you could, I don't know how to program AI, but I assume you can say to Claude, every answer you give me on every company has to be different from the consensus.

Or the opposite of consensus. Or, well, the opposite is pretty categorical, but let's say has to deviate from the consensus. And no consensus thinking is welcome here.

You can say that. You can make it be different. Can you make it be better? Because, you see, this is the catch.

Because the people you compete against in the investment arena are also intelligent, educated, literate, computer able, highly motivated. They're pretty good. Much of the time, they're as close to right as you can get.

So if you believe the consensus is as close to right as most people can get, then if you tell Claude, I only want non-consensus thinking, then it has a high probability of being wrong. That's not a good thing. In a way, AI in investing could work if everyone in investing uses AI.

No, I don't think so. Because if everybody uses AI, and let's assume everybody's AI is the same. Not same, different.

Well, are they? That's the question. Maybe that's what my partner Bruce Karsh, the recovered lawyer, would call a gating question. Are some AI models smarter than others? Since they all have super IQs, computer power, and they all are trained on the same history, are some smarter than others? I don't know.

I need somebody to tell me. But if they're all equally smart, and you say to them, this is an interesting thing. So they're all equally smart.

Everyone says to all of them, write out a process that will produce investment success. Theoretically, they all write the same one, because they are all equally smart and have the same training. And if they all follow it perfectly, they all produce the same result.

So there's no superiority. But then everybody makes 10%, nobody makes 8 or 12. Right.

But then what do you get paid for making 10%? But AI doesn't need to be paid. Yeah, but who employs AI? That person wants to get paid. Nobody's employing AI for fun, right? They're employing AI because they want to get rich.

And getting rich and investing comes from superiority, not from averageness. How do you get superiority? I don't want to ask you this question, but... You can choose not to answer it. Yes.

Do you think the world is going to socialism and AI is triggering that? I don't consider myself a futurist. I don't engage in that kind of... That kind of thinking is what we call blue sky. And I am partial to what Albert Einstein said on the subject.

He said something like, I don't think about the future, it'll come soon enough. But do I think about the past? I think about the past to learn from it. And I try to... I'm great on these quotes.

And my favorite on this subject is from Mark Twain, who supposedly, but I think didn't really say it, supposedly said, history does not repeat, but it does rhyme. There are certain themes that rhyme from instance to instance in history. And they mostly have to do with human nature.

And I think human nature doesn't change incredibly slowly. I mean, when we talk about fight or flight, people start talking about the Stone Age man and the watering hole, right? Well, that was millions of years ago. These things are still relevant.

So the point is that I think you can learn from history. And what you mostly learn are the lessons of human nature. Mark Twain said that.

And Napoleon said history is written by the winners of tomorrow. So we don't even know how accurate the history we have access to really is. Well, except for we have the numbers.

Financial history. Yes, financially. When you have numerical history, it's probably not that subjective.

But maybe not the psychology that led to the numbers. Right, right. Well, I mean, I just read Andrew Ross Sorkin's new book, 1929.

It's a great book. Yeah, I like it. I have a memo coming out this week talking about private credit.

And I refer to it in that. What's happening with private credit? Let me just finish this thought, okay? So, you know, he explains the events that led up to the Great Crash rather vividly. He gives you a picture.

And I think there's a lot to be learned from that. And I try to convey some of the lessons of 1929 in this new memo, which is coming out in 2026. I hope that the history that Andrew describes is the real history.

Because if not, the lessons will not be that valid. That's all I can say. Any last words, Howard, to our audience? Any piece of advice? Well, what I always say to people about investing, Nikhil, is that investing is a puzzle.

Challenging puzzle. Investing, I define as positioning your capital to take advantage of future developments. But I also say that future developments can't be predicted accurately.

So you need superior insight to get it more right than most people. But you're not going to get it right all the time. If you have to be right all the time, if there's something in your makeup that says you're going to be unhappy if you're not right all the time, don't become an investor.

Taleb, in his book Fooled by Randomness, talks about the difference between investing and dentistry. He says if you go to dental school and you learn how to fill a cavity, and you fill the cavity that way every time, you'll be successful every time. So if you have to be successful every time, become a dentist.

Or an engineer, a civil engineer. You say, I want to build a bridge from here to there. He does his calculations.

How much steel? How much concrete? Every time, the bridge stands. But it's not that interesting, in my opinion. And so investing deals with this fascinating puzzle that you're trying to solve to a better degree than others, even though there are no laws that work absolutely.

Fascinating. And I love it. And it's still a puzzle.

It's still a puzzle I'm trying to peel like the onion. But you're never done. And that makes it interesting.

And so that's the kind of person who should be engaged in it. It's interesting that you resonate with Taleb, because somehow when I read your stuff and his stuff, I find you guys on the opposite ends of the spectrum. I think Taleb, when he ran his fund, he was buying way out of the money,

call options and put options, hoping for black swan events.

And you, in a way, are doing the opposite of that. When you're buying debt in distress, hoping that the percentage of things being black swans are lower than what the market perceives. Right.

In that sense, I think we are different. I mean, I said earlier, I believe in putting together a portfolio that would do well in the main portion of the probability distribution. He's advocating protection that can be bought cheaply because it only deals with the tail event, which rarely happens.

So people underestimate the value of the tail protection. But so you can buy, you can cheaply buy protection against something that is unlikely to happen, which in theory is a good thing. But since it happens so rarely, that the value of that underpriced tail protection doesn't become apparent very often.

Even he compares in the book, as I recall, I only read it 22 years ago. I read all his books. I love him.

Yeah. But even he says that investing is like Russian roulette. But rather than having a bullet in the six chambers, there are 100 chambers in the gun.

So the bullet rarely goes off. And I don't think you can, I mean, so you could spend 1% of your money to buy tail protection every year. But I think it's, I think if that's going to be your whole existence, it's going to be a dull existence because the explosion will happen so infrequently.

And he also said that even COVID was not an explosion. He said it's not a black swan event. So if COVID wasn't, the question is, what is? Well, it's, yes, I mean, you know, obviously COVID was not the first pandemic, but it was, I would say it was the first pandemic of the modern era.

And it was the first pandemic that encountered the digital economy and many, many, and globalization, many, many other things. So it was kind of unique or certainly innovative in a lot of ways. And I would guess we could find, we could say some people dealt with it better and worse, which is, to me, the definition of what makes something interesting, the ability to deal with it better or worse.

Thank you, Howard, for taking the time today. This was incredibly interesting. And I feel like I'm learning so much from you.

Well, and I have learned from speaking with you and I've enjoyed it. Your questions were, in many cases, novel and provocative. Thank you.

Thank you for doing this.